

CIE

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TITLE: FitTrack: Personalized Fitness and Medicine Management System

SCHOOL: Chitkara University, Rajpura, Punjab

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GOVT PORTAL NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS**

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ABSTRACT

FitTrack is a web-based fitness and health management application designed to assist users in achieving their physical goals through intelligent workout scheduling and comprehensive routine tracking. The system integrates personalized fitness planning with medicine reminder functionality, creating a unified platform that supports both physical wellness and health adherence. Developed using React for the frontend and MongoDB for the backend, FitTrack offers a responsive and user-friendly interface combined with efficient data management.

The primary objective of FitTrack is to generate customized workout schedules based on individual user goals, such as weight loss, muscle gain, endurance improvement, or general fitness maintenance. By considering user-defined preferences, fitness levels, and targets, the application provides structured and goal-oriented exercise plans. Additionally, FitTrack enables users to monitor daily fitness activities, including completed workouts, calories burned, steps taken, and sleep patterns, promoting self-awareness and consistency.

A distinguishing feature of the system is its medicine tracking module, which allows users to input medication details, dosage schedules, and reminder timings. Automated alerts help ensure timely intake, addressing a common challenge in maintaining treatment adherence. This integration of fitness and medication management enhances the practical value of the application, particularly for users managing health conditions alongside fitness objectives.

FitTrack emphasizes usability, data accuracy, and motivation through progress visualization and routine consistency tracking. By combining exercise planning, activity monitoring, and medicine reminders within a single platform, the system aims to encourage healthier lifestyles, improve discipline, and support long-term wellness. The project demonstrates the application of modern web technologies in solving real-world health and fitness challenges, offering a scalable foundation for future enhancements such as AI-driven recommendations and wearable device integration.

1.1 Problem Background

Maintaining a healthy lifestyle requires consistent workout planning, routine tracking, and timely medicine intake, which many individuals struggle to manage effectively. Existing fitness applications often focus solely on activity monitoring, while medication reminder apps operate separately, leading to fragmented health management. This lack of integration results in poor adherence to fitness goals and medical schedules. Users need a unified system that simplifies workout scheduling, tracks daily progress, and ensures medication compliance, promoting discipline, organization, and overall well-being through a single, user-friendly platform.

1.2 Objective of the Project

The objective of FitTrack is to provide a unified platform that generates personalized workout schedules, tracks daily fitness activities, and manages medicine reminders, enabling users to maintain consistency, improve health discipline, and efficiently monitor progress toward their physical and wellness goals .

1.3 Proposed Solution / System Overview

FitTrack is proposed as an integrated fitness and health management system designed to deliver personalized workout scheduling and medicine tracking within a single platform. The system functions by collecting user inputs such as physical goals, fitness preferences, and medication details, then generating structured workout plans and timely reminders. The major components of the system include the user management module, workout scheduling module, activity tracking dashboard, and medicine reminder module. Additionally, the system maintains secure data storage and user-specific records, supporting clear feature definition, originality documentation, and ownership clarity for copyright purposes.

1.4 Key Features or Functionality

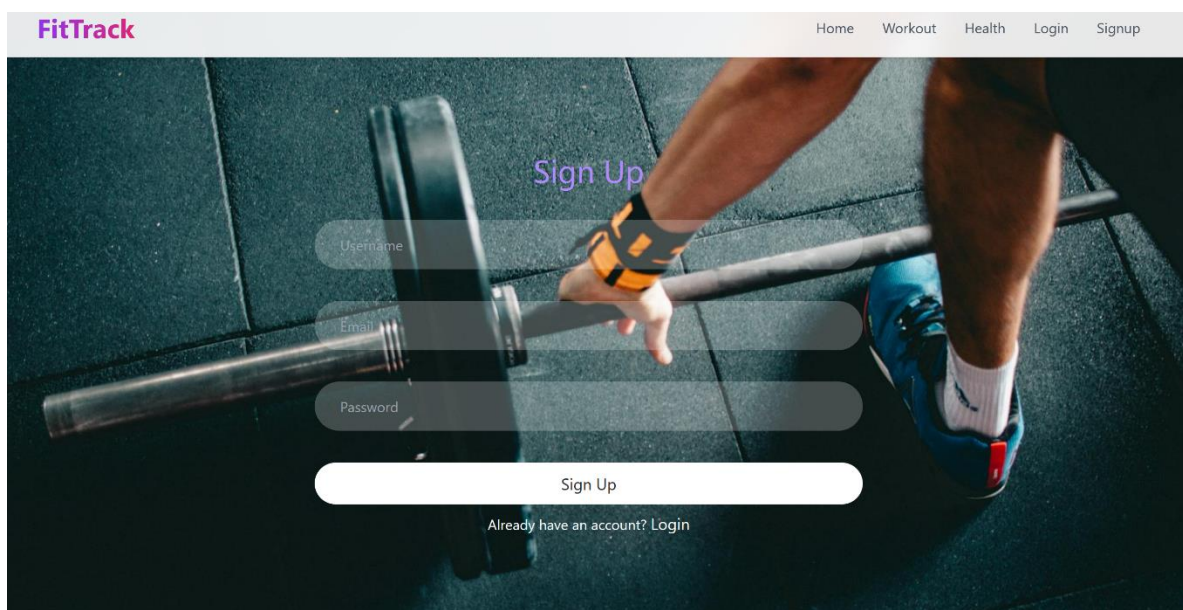
- **User Authentication & Profile Management:** provides secure login, user profiles, and safe storage of personal fitness data.
- **Personalized Workout Scheduling:** generates customized workout plans based on user-defined fitness goals for structured training.
- **Activity Tracking & Monitoring:** tracks workouts, calories burned, steps taken, and overall routine consistency.
- **Progress Tracking & Dashboard:** displays fitness progress through an interactive dashboard with visual insights and history management.
- **Medicine Tracker:** allows users to input medication details and reminder timings, with browser notifications to ensure timely dose intake and streak tracking for consistency.

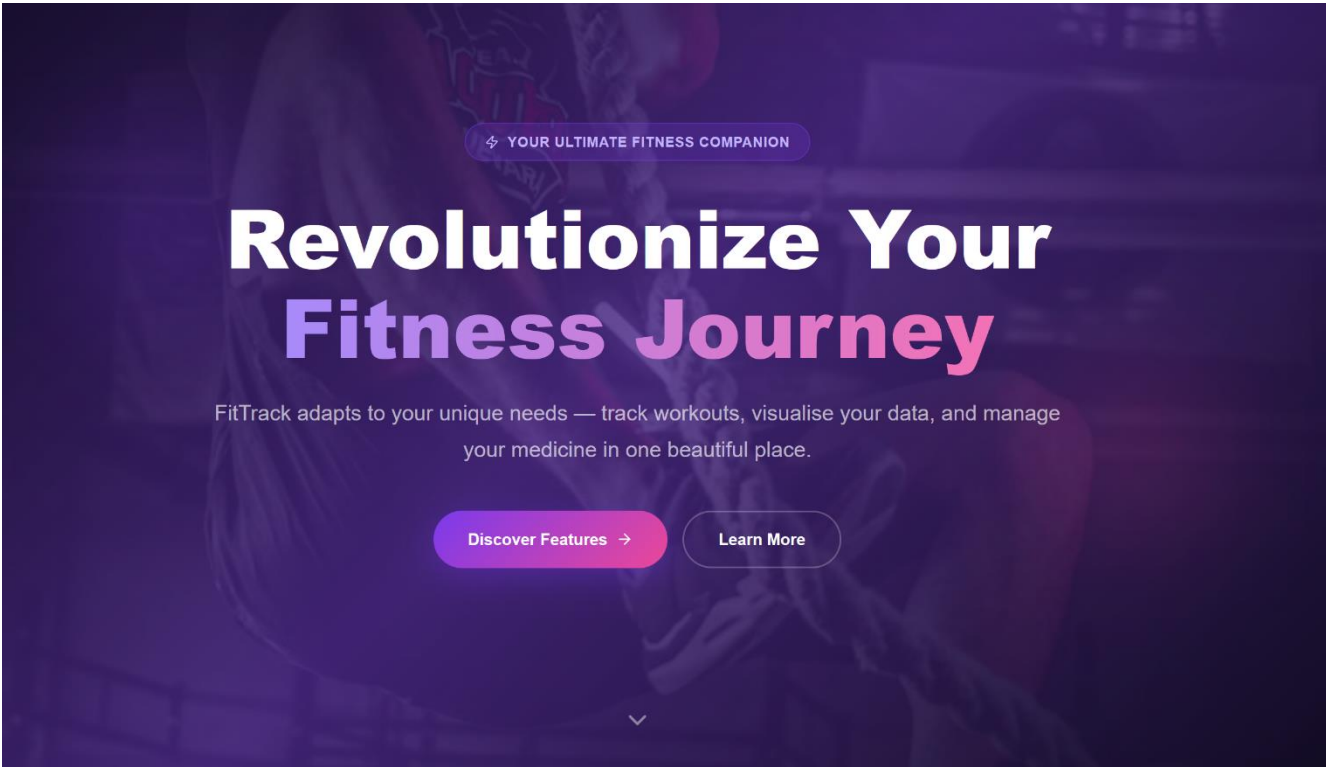
1.5 Technology / Method Used

- **Frontend Development:** built using React.js with responsive UI design and integration of charts (graphs and pie charts) for data visualization.
- **Backend Services:** developed with Node.js and Express.js to handle server-side logic and RESTful APIs.
- **Database Management:** MongoDB is used for efficient and scalable data storage.
- **Authentication & Security:** Session-based authentication is implemented using localStorage to maintain user login state across the application.
- **API Integration:** RESTful APIs are used for seamless communication between frontend and backend.
- **Data Visualization:** libraries like Chart.js or Recharts are used in React to represent fitness data through graphs and pie charts.

1.6 Expected Outcome & Benefits

FitTrack is expected to improve users' consistency in workouts and medication adherence through structured scheduling and reminders. The system enhances self-monitoring, promotes healthier lifestyle habits, and simplifies routine management. Users benefit from better organization, progress visibility, and convenience, while the application demonstrates practical integration of fitness and health management technologies.





Your Fitness Journey

Step 1 of 5

What is your current fitness level?

Beginner
Just getting started

Intermediate
Some experience

Advanced
Training regularly

Continue

Your Fitness Journey

Dashboard

Exercises

Achievements

Dashboard — Welcome, test3! 🙌

04/26/2026



Mon, Apr 20

Tue, Apr 21

Wed, Apr 22

Thu, Apr 23

Fri, Apr 24

Sat, Apr 25

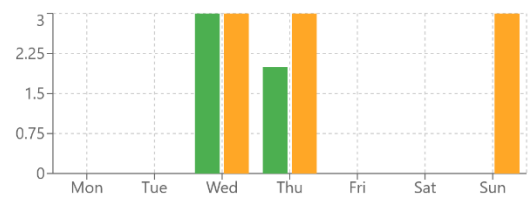
Sun, Apr 26



Today's Progress



Weekly Overview



Dashboard

Exercises

Achievements

Your Exercise Routine

1 of 3 exercises done today

33%

Pending

2 left



Walking

3 pts on completion

Done



Light Yoga

4 pts on completion

Done

Completed

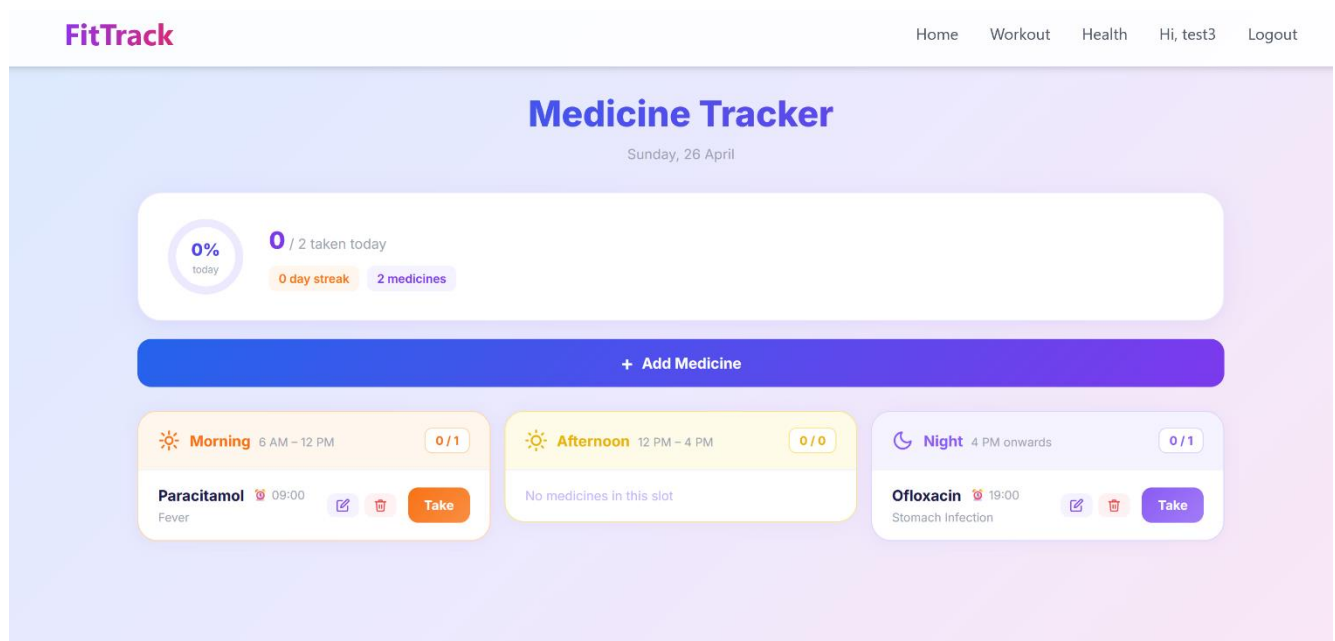
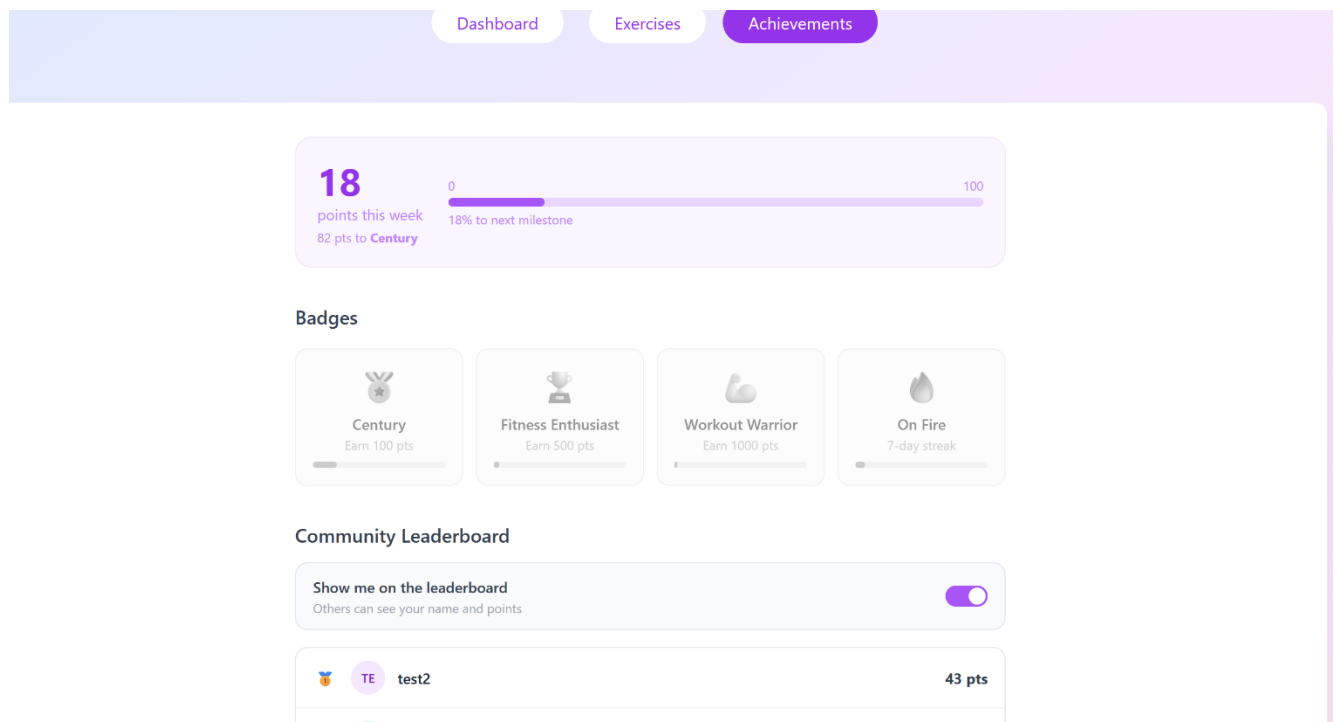
1 done



Stretching

+2 pts earned

Done



1.7 Original Contribution Statement

FitTrack presents a novel approach by combining fitness planning, routine tracking, and medication adherence into a unified, user-centric health management system.